

HAYA HEALTH CARE

"Predict Today, Protect Tomorrow"

PROJECT TYPE:

AI-Powered Multi-Disease Prediction and Healthcare Analytics Platform

FOUNDER & LEAD ENGINEER:

Hariprasath L

AI & Data Science Student | Python Full Stack Developer

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1. Executive Summary

Haya Health Care is an advanced medical software platform designed to optimize disease risk assessment and patient health data profiling. By integrating statistical Machine Learning classifiers with a highly responsive web layout, Haya enables clinical practitioners, doctors, and healthcare administrators to predict high-risk status indexes for multiple disease vectors immediately. The system facilitates personalized analytics trends, secure patient records registry, and natural-voice interface operations to maximize medical documentation speed and diagnostic accuracies.

2. Technical Architecture & Tech Stack

Layer	Technologies Utilized
Frontend Core	React.js, Tailwind CSS (v4), Bootstrap, Material UI
Visualizations	Chart.js, React-Chartjs-2
Backend API	Python, FastAPI, Uvicorn, CORS Middleware
AI & Machine Learning	Scikit-Learn, TensorFlow, XGBoost, Pandas, NumPy
Database Structures	MongoDB (NoSQL), MySQL (RDBMS)
Dev Environment	VS Code workspace tasks & launch debuggers
Inference Utilities	HTML5 Speech Recognition (Voice input), Text-to-Speech (Voice out)

3. Core Platform Modules & Features

- Multi-Disease Diagnostic Pipelines

Includes 7 clinical classifiers covering Kidney Disease, Diabetes, Heart Disease, Liver Disease, Parkinson's Disease, Lung Cancer, and Stroke risk prediction. All parameters are evaluated via simulated Decision Bounds mirroring ML thresholds.

- Dynamic Admin Control Deck

Allows administrators to modify founder details, upload founder profile photos, manage active doctors directory, edit patient registrations, and trigger model retrains dynamically with automatic LocalStorage data persistence.

- Symptom & AI Chatbot Companion

A diagnostic chatbot leveraging hybrid LLM endpoints to check symptoms, suggest dietary guidelines, formulate exercise workouts, and trigger critical medical alerts for emergency parameters.

- Dataset Upload Profiler & Download Center

Allows doctors to drag-and-drop clinical CSV/Excel files to calculate data structures (row counts, missing values, duplicates) and generate summary statistics. Provides export hubs for CSV data downloads.

- Voice Control & Accessibility Integration

Includes speech synthesis (voice readings of predictions) and web speech recognition (voice commands to autofill form values) for enhanced accessibility.

4. Founder & Lead Architect Profile

Hariprasath L

AI Engineer | Full Stack Developer | Machine Learning Enthusiast

Hariprasath L is a dedicated developer with deep capabilities in designing statistical Machine Learning pipelines and scalable web client-server systems. With expertise in Python (Django, FastAPI), React, and SQL/NoSQL databases, Hariprasath leads the architecture of Haya Health Care, creating secure medical diagnostic networks designed to optimize clinical operations and increase diagnostic speeds.

Key Technical Skillsets:

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|----------------------|---|
| - AI & ML: | Machine Learning, Deep Learning, Data Analytics, NLP, Computer Vision |
| - Backend Stack: | Python, Django, FastAPI |
| - Frontend Dev: | React.js, Tailwind CSS, Bootstrap, JavaScript, HTML5, CSS3 |
| - Databases: | MySQL, MongoDB |
| - Tools & Platforms: | Git & GitHub, VS Code, Jupyter, Google Colab, Streamlit |

5. App User Interfaces & Visualizations

Below are high-level
prediction panels



Figure 5.1: Haya Health Care Analytics Dashboard UI



Figure 5.2: Kidney Disease Parameter Assessment UI